Measurement protocol: CPU swap test AT32UC3A0512

Oct 19, 2026 · @Pavel Miksicek

Document data and approval

The protocol is approved before step 1 begins. Approval: the operator uploads the PDF of this version and its SHA-256 hash to the measurement repository (folder PROTOCOL_v1.0, commit and push). The approved version is not changed afterwards; any change means a new version with a stated reason. During the test, only the fields intended for it are filled in: measurement dates, the approval, the table of records and the section Results and conclusion.

Item	Value
Subject	swap of the AT32UC3A0512 CPU (silicon revision L) between boards PCB1 and PCB2 using the A-B-A method
Protocol version	1.0
Storage	private repository METEL-EU/at32uc3a0512-swaptest on GitHub.com
SHA-256 of the approved version	kept outside the document: hash of the PDF of the approved version in the SHA256SUMS file in the measurement repository and in the table of records of the completed protocol
Performed by	Pavel Mikšíček
Checked by (second signature)	Milan Holinka
Approval	[to be filled in: date and commit of the approval]
Measurement dates	[to be filled in: dates of steps 1, 6 and 7]

Purpose and scope

The test determines whether the flash read failure follows the CPU or the printed circuit board (hereafter board, PCB) on which the CPU is mounted and powered. Exchanges of complete boards so far only show that board PCB1 fails as a whole; the CPU itself is

therefore resoldered.

The procedure follows the A-B-A method described in the Microchip article SWAP TEST: How to determine if a Microchip part is causing the board failure:

1. Clean board PCB1 (with CPU-X) and board PCB2 (with CPU-Y) with a brush and flux remover. Test repeatedly and confirm that PCB1 fails and PCB2 is good.
2. ~~Remove CPU-Y from PCB2, solder it back onto the same board and test. PCB2 must remain good.~~ Not performed.
3. ~~Remove CPU-X from PCB1, solder it back onto the same board and test. PCB1 must still fail. This rules out a soldering defect.~~ Not performed.
4. Remove CPU-Y from PCB2 and CPU-X from PCB1.
5. Solder CPU-Y onto PCB1 and CPU-X onto PCB2.
6. Test. If the CPU is defective, PCB2 (originally good) fails and PCB1 (originally failing) passes.
7. Remove both CPUs, return them to their original boards and test. PCB1 must fail again and PCB2 must pass.

Steps 1 and 4–7 are performed. Steps 2 and 3 (control resoldering of each CPU onto its own board) are omitted because step 6 rules out a soldering defect: if CPU-X also fails when freshly soldered onto PCB2, the original joint on PCB1 cannot be the cause. Step 1 is measured on the boards as delivered: both boards were professionally cleaned in production after assembly and no soldering has been done on them since; only the laser label was added. They are therefore not cleaned again. The repeated testing required by the article is provided by the regression run in the preparation phase and the measurement in step 1.

The test shows whether the defect lies in the CPU or in its surroundings on the board (crystal, power supply, passive components, SPI flash, solder joints). It does not show the cause inside the chip. The other parts of the measurement setup are the same in all steps: power supply, USB–RS-485 converter with a floating ground on a short line, programmer, oscilloscope and measurement PC (see Equipment and tools).

Units under test

The pairs PCB1 + CPU-X (originally failing) and PCB2 + CPU-Y (originally good) are tested; both CPUs are silicon revision L. The electronic identifiers that the diagnostics prints in every run are decisive; the short labels serve the text of the protocol, the photographs and the external company.

Short label	Decisive identifier	Additional identifier	Physical marking
PCB1 (originally failing)	Board ID 8a32128415213248	the board has no serial number	laser label PCB1, photographs
CPU-X (originally on PCB1)	SN 141916cb4450334d342e3120ff060b	JTAG ID 0xB1EDC03F (revision L), marking	none
PCB2 (originally good)	Board ID 8a3203841518175d	the board has no serial number	laser label PCB2, photographs
CPU-Y (originally on PCB2)	SN ba16905e4450334d342e3120ff0714	JTAG ID 0xB1EDC03F (revision L), marking	none

- **CPU SN** is the 120-bit number from the factory page (0x80800204–0x80800212) written by the chip manufacturer; it is unique for each chip. The JTAG ID and the marking are not unique and only show the same revision and lot.
- **Board ID** is the 64-bit UID of the SPI flash chip on the board. It identifies this chip, not the board itself; it is valid because the SPI flash is not resoldered during the test.
- **Board label** PCB1/PCB2 is laser-marked on a free area of the board with no tracks under the solder mask. In the photographs it is in the bottom right corner of the board.
- **Physical marking of the CPU** is not used because a marker does not adhere to the package. CPU-X and CPU-Y are distinguished by the SN only.

Validity rule: a run is valid only if the CPU SN and the Board ID in the output match the expected pair – in steps 1 and 7 PCB1 + CPU-X and PCB2 + CPU-Y, in step 6 PCB1 + CPU-Y and PCB2 + CPU-X.

CPU marking (identical on CPU-X and CPU-Y, readable in the photographs of the records): ATMEL / 32UC3A0512-U / 2617L PH / 2617QR4.

The initial state of both pairs is documented by step 1 with the final firmware.

Diagnostic firmware and definition of an error

An error is a read value other than 0xFFFFFFFF. The diagnostics is a standalone firmware in the CPU; after start-up it performs 100 cycles of 480 reads and compares each with this constant. The final firmware version 2 (folder 20261010_swaptest_final, description FINAL.md) completed a regression run on PCB1 and PCB2 on 2026-10-15: PCB1 + CPU-X FAIL, PCB2 + CPU-Y PASS, consistent with the original state (see Preparation).

- **What it measures:** 24 tables (A–X) × 20 copies of the check function at fixed addresses in the range 0x8000A000–0x8007FC03. Each function reads a word with the value 0xFFFFFFFF through a computed address (instructions bfextu, add, ld.w).
- **Origin:** the sequence was obtained by minimising AES code that failed in the product. Some CPUs of this type have shown errors in AES encryption in the product over the long term (several years).
- **Reference reads:** counted separately to distinguish a failure of the reference from a failure of the check read (FINAL.md, section 1).
- **UART output (115200 Bd, 8N1):** identification block at the start and end of the run; in the 1st cycle every mismatch with its address and read value; per-cell summary with the error count and failing bits; machine-readable RESULT line.
- **Identification block:** CPU SN, JTAG ID, Board ID, fuses, FLASHC registers, clock settings, reset cause, BOD state, CRC32 of the flash image computed at run time, planned number of cycles.
- **Writes:** the firmware only reads from memory. The only write sets 1 wait state in FLASHC.FCR; bit 8 of FCR is not changed.

Item	Value
Source code	https://github.com/METEL-EU/at32uc3a0512-swaptest/tree/ab717731ff772f467183682d3439bbcb72d3caf9/20261010_swaptest_final
Compiler	WinAVR 20100110 (avr32-gcc 4.3.2), GNU Make 3.81
Build	make clean all
SHA-256 .bin	2ECFE5317649FABD27797608F3F9924AAAC45B4DAA870A7546195E9975294440 (version 2, CRC32 0xfb47ddb9)
SHA-256 .elf	6675E8E3E4B4FF880ADD0A8B817B13226083ECCD2A64DC31F07DB1B353E08909
Reproducibility	two independent builds are bit-identical (FINAL.md, section 7.1)

Item	Value
Test core	24 tables and 480 check functions at fixed addresses 0x8000A000–0x8007FC01 according to placement.ld (verify_build.py, FINAL.md, sections 1.1 and 7.2)

Operating conditions and their evidence

The CPU runs at 64 MHz with 1 flash wait state, i.e. within the limits of datasheet doc32058K. The voltages and the temperature are measured during the 1st run of every configuration and entered in the configuration record; for comparison, the Evidence column gives the values from the regression run (records REG_PCB1-X and REG_PCB2-Y).

Quantity	Value	Limit	Evidence
OSC0 crystal	8 MHz	0.45–16 MHz (doc32058K, Table 38-15)	indirect: the cycle length by the PC clock in the regression run, 1.118 s (PCB1) and 1.115 s (PCB2), corresponds to 64 MHz (FINAL.md, section 2.3), i.e. to an 8 MHz crystal
fVCO (PLL0)	128 MHz	80–180 MHz	PLL0 register in the output
CPU, HSB, PBB	64 MHz	max. 66 MHz	clock registers in the output
PBA	32 MHz	max. 66 MHz	clock registers in the output
Flash wait states	1	1 above 33 MHz	FLASHC.FCR in the output
Crystal start-up time	2048 RC cycles ≈ 17.8 ms (OSCCTRL0.STARTUP = 3)	typ. 4 ms for an 8 MHz crystal (doc32058K, Table 38-15); STARTUP = 3 is 2048 RC cycles ≈ 18 ms (Table 13-7)	OSCCTRL0 in the output

Quantity	Value	Limit	Evidence
VDDIO	= VDDIN (connected on the board)	3.0–3.6 V	oscilloscope, during the run
VDDIN	measured during the 1st run of every configuration	3.0–3.6 V	oscilloscope; regression run 3.31 V (PCB1 and PCB2)
VDDCORE	measured during the 1st run of every configuration	1.65–1.95 V	oscilloscope; regression run 1.77 V (PCB1), 1.81 V (PCB2)
Ambient temperature	measured during the 1st run of every configuration	–	thermometer; regression run 24.6–25.1 °C
Fuses	0xFFFFFFFF / 0xFFFFFFFF	supported configuration	output and JTAG
BOD	disabled (BODEN = 3)	–	PM.BOD in the output
Security bit	not set	–	atprogram info

Errata: doc32058K, section 41.1 (revisions K, L, M) lists a single flash-related item, FLASHC 1: reading from the flash may fail after a fuse write operation (commands LP, UP, WGPB, EGPB, SSB, PGPFB, EAGPF). The firmware issues none of these commands and the runs start at power-on. The errata state nothing about prefetch or wait states. Item Power Manager 1 (BOD level set above VDDCORE) does not apply; the BOD is disabled. The separate errata DS80000980A from 2021 add only an item on flash erase verification, which is not related to the test.

Equipment and tools

The same equipment is used in all steps; any replacement is recorded.

Item	Identification
Programmer	Atmel JTAGICE3, S/N J30200000214, firmware 3.38
Programming tool	atprogram 6.2.1127.0 from Atmel Studio 7 (C:\Program Files (x86)\Atmel\Studio\7.0\atbackend), run with TEMP=C:\Windows\Temp

Item	Identification
Power supply	48 V, Mean Well SDR-120-48
USB–RS-485 converter	METEL USB485-BOX, floating ground, short line, 115200 Bd, 8N1
Capture script	tools/capture.ps1 and the other scripts in tools/ of the measurement repository; version = git commit (README.md)
Oscilloscope	Keysight DSO-X 2024A, S/N MY52160502, 10:1 probes, DC coupling; not calibrated
Thermometer	indicative instrument, manufacturer METEL
Camera	Nikon D3400, time stamp in the image; CPU marking and board label readable in the pictures

Notes:

- The VTref pin of the programmer is not connected on the board. The Target voltage reported by atprogram therefore does not correspond to the board supply and is not used; the flash content is verified by readback. The JTAG is disconnected during the runs.
- The script prog3_0512.bat does not work on the measurement PC because its atprogram lacks the UC3 device descriptions. The commands given in the procedure apply.
- The command verify --format bin fails in this version of atprogram; the flash content is verified by readback and hash comparison.

Definition of a run and evaluation of a configuration

Each configuration is measured with three runs of 100 cycles, always after power-on with the JTAG disconnected.

Run: switch off the power, wait at least 10 s, switch on, record the output up to the RESULT line, switch off. A run takes ≈ 113 s for a good pair and ≈ 120 s for a failing pair (regression run: 112.8 s and 120.3 s).

Run result	Condition
FAIL	RESULT line with an error count ⩾ 1
PASS	RESULT line, 100 cycles completed, 0 errors

Run result	Condition
INCOMPLETE	no RESULT line within 142 s (FINAL.md: 100 × 1.12 s + 30 s) of power-on
DID NOT START	no output within 5 s of power-on

- A run is valid only if the output contains the CPU SN and the Board ID of the expected pair for the step (see Units under test) and a CRC32 of the flash image equal to that of the binary.
- An INCOMPLETE, DID NOT START or invalid run counts neither as PASS nor as FAIL. The run is repeated and the number of attempts is recorded. After 3 such attempts in a row, the measurement is stopped and the cause is investigated.
- **Configuration result:** FAIL = 3 of 3 valid runs FAIL; PASS = 3 of 3 valid runs PASS. A mixed result = INCONSISTENT; the test is stopped and evaluated before the next step.

Measurement procedure for one configuration

This list is carried out for every board + CPU pair in every measurement step. The items are ticked in the configuration record, except the last one: the hash is entered in the table of records only after the hashes have been computed.

- [ ] Connect the JTAG and read back the entire flash, the user page and the FLASHC registers. The SHA-256 of the range 0x80000000–0x8007FC03 must match the binary. Mismatch = stop and record. Script tools/readback.ps1 (README.md).
- [ ] Disconnect the JTAG from the board.
- [ ] Photograph the board with the CPU so that the CPU marking and the board label are readable. The photograph has a time stamp in the image and is taken during the measurement of the configuration, between the readback and the end of the UART capture.
- [ ] Start the UART capture into a new file (script tools/capture.ps1).
- [ ] Perform three runs according to the definition of a run.
- [ ] During the first run, measure VDDIN and VDDCORE (VDDIO is connected to VDDIN on the board) and the ambient temperature and record the values. The oscilloscope probes are connected with the power off; the screenshot is saved within the first minute of the run.
- [ ] Stop the capture, check the CPU SN, the Board ID, the image CRC32 and the RESULT lines in the output (script tools/evaluate.py) and enter the results in the configuration record.

- [ ] Compute the SHA-256 hashes into the SHA256SUMS file of the configuration (script tools/sha256sums.ps1), commit the revision and push it to GitHub.
- [ ] Enter the hash of the SHA256SUMS file in the table of records.

Test sequence by phase

The test has a preparation phase, three measurement steps (1, 6 and 7) and two rounds of resoldering; the step numbering follows the Microchip article. After step 1 the test continues only if the failure was reproduced.

1. **Preparation**
   - Mark the boards with the laser labels PCB1 and PCB2.
   - Program the final binary into CPU-X and CPU-Y (chiperase, program) and verify it by readback.
   - Document the regression run of the final firmware on PCB1 and PCB2 with their CPUs. Results: folders REG_PCB1-X (FAIL, 3 valid runs of 3) and REG_PCB2-Y (PASS, 3 valid runs of 3) in the repository METEL-EU/at32uc3a0512-swaptest, see the table of records.
   - Approve the protocol: upload the PDF of this version and its hash to the measurement repository (PROTOCOL_v1.0, commit and push); the source code, the binary and the .elf and .map files are already in the repository (commit ab71773). Step 1 starts only after that.
2. **Step 1** – measure PCB1 + CPU-X and PCB2 + CPU-Y as delivered, without further cleaning (see Purpose and scope). The test continues only with the result FAIL / PASS; otherwise the test ends and nothing is soldered.
3. **Resoldering 1 (steps 4 and 5)** – CPU-X onto PCB2, CPU-Y onto PCB1.
4. **Step 6** – measure PCB1 + CPU-Y and PCB2 + CPU-X.
5. **Resoldering 2 and step 7** – both CPUs back onto their original boards, measure PCB1 + CPU-X and PCB2 + CPU-Y.
6. **Evaluation** according to the interpretation table and writing of the conclusion.

After each measurement step, the records and their hashes are immediately uploaded to the measurement repository.

After resoldering, the CPUs are not programmed again. The firmware is in the internal flash and moves with the chip; the readback before each configuration shows that the content has not changed.

Each CPU went through one reflow during board production; the test adds four thermal cycles (removal in step 4, soldering in step 5, removal and soldering in step 7). The datasheet allows at most three reflow passes per component (doc32058K, section 39.3).

In step 6 each CPU is therefore exactly at this limit, and in step 7 it exceeds it by two reflows. The number of thermal cycles follows from the A-B-A method of the Microchip article. Baking before each removal is mandatory according to the requirements for the external company.

Evaluation of results

The conclusion is determined solely by this table, written before the start of the test; the criterion for a defective CPU is taken from the Microchip article. X and Y denote CPU-X and CPU-Y; the results are given as "first pair / second pair".

Step 1: PCB1+X / PCB2+Y	Step 6: PCB1+Y / PCB2+X	Step 7: PCB1+X / PCB2+Y	Conclusion
other than FAIL / PASS	–	–	The failure was not reproduced; the test ends, nothing is soldered
FAIL / PASS	PASS / FAIL	FAIL / PASS	**Defective CPU-X**: complete swap in step 6, same state in step 7
FAIL / PASS	PASS / FAIL	other than FAIL / PASS	According to the article: the cause is in another part of the board
FAIL / PASS	FAIL / PASS	FAIL / PASS	The defect follows board PCB1
FAIL / PASS	PASS / PASS	PASS / PASS	Soldering defect of CPU-X on board PCB1
FAIL / PASS	PASS / PASS	FAIL / PASS	Interaction of CPU-X and board PCB1; neither alone is sufficient
FAIL / PASS	other combination	other combination	Inconclusive

A configuration with the result INCONSISTENT makes the whole test inconclusive. Step 7 is performed above the limit of three reflows (doc32058K, section 39.3); for conclusions that rely on the result of step 7, an influence of the thermal stress of the CPU during resoldering therefore cannot be ruled out.

Records and their protection

Records are the files created during the measurement; their SHA-256 hashes are uploaded to the private repository on GitHub.com immediately after each measurement step. A later change to any file would change its hash.

- **What a record is:** the raw diagnostic output from the UART captured by a script on the PC, the same output with time stamps, readbacks of the flash, the user page and FLASHC via JTAG, photographs, measured voltages and temperatures.
- **Storage:** one folder per step and configuration, e.g. STEP1_PCB1-X. Repository: METEL-EU/at32uc3a0512-swaptest.
- **Hashes:** after each step a SHA256SUMS file with the hash of every file, then a commit and a push to GitHub. Raw records are not edited; a correction is made only as a new file with a note.
- **Sending:** the hash of the SHA256SUMS file is entered in the table of records after each step. After the test is completed, the filled-in and signed protocol is sent by e-mail to the recipient of the results; the copy held by the recipient is evidence that does not depend on the repository administrator.
- **Repository:** a private organisation repository because of the confidentiality of the source code; the recipient of the results can be given read access. The repository does not technically prevent rewriting of the history; the evidential value of the records rests on the SHA-256 hashes in the table of records and on the copy of the signed protocol held by the recipient of the results.
- **Measurement outputs:**
  - the approved protocol;
  - the source code, the binary, the .elf and .map files and their hashes;
  - all UART records and SHA256SUMS files;
  - the memory readbacks;
  - photographs and resoldering reports;
  - the table of results and the conclusion according to the interpretation table.

Table of records: the hash of the SHA256SUMS file of each record folder. The signatures of the operator and the checker on the completed protocol also confirm this table.

Record	Folder	Commit	SHA-256 of the SHA256SUMS file
Regression run, PCB1 + CPU-X, 1st attempt, superseded: the board did not yet carry the laser label	REG_PCB1-X_attempt1	d410b5d	198C38DB349085A949D9A04B053EF86BAC647D450A0AF3E5005F78136C3FE362

Record	Folder	Commit	SHA-256 of the SHA256SUMS file
Regression run, PCB1 + CPU-X	REG_PCB1-X	8c2a499	45F8A367935539E0AAF34262BF8E65CEB28A48D DACAEB3DB4574EDC32DF998EC
Regression run, PCB2 + CPU-Y	REG_PCB2-Y	6a933f3	23F536FAD2D0DF9523C550FD7C7E6149B1B08B8 AD3B4BCCDBBF936A284F36A3F
Approved protocol, version 1.0	PROTOCOL_v1.0	[to be filled in]	[to be filled in]
Step 1, PCB1 + CPU-X	[to be filled in]	[to be filled in]	[to be filled in]
Step 1, PCB2 + CPU-Y	[to be filled in]	[to be filled in]	[to be filled in]
Step 6, PCB1 + CPU-Y	[to be filled in]	[to be filled in]	[to be filled in]
Step 6, PCB2 + CPU-X	[to be filled in]	[to be filled in]	[to be filled in]
Step 7, PCB1 + CPU-X	[to be filled in]	[to be filled in]	[to be filled in]
Step 7, PCB2 + CPU-Y	[to be filled in]	[to be filled in]	[to be filled in]

Requirements for resoldering by the external company

The external company receives a separate work order; the protocol requires the records listed below from it. Package: LQFP-144.

- **Baking:** the CPU has moisture sensitivity level MSL 3 according to J-STD-020D (doc32058K, Table 39-6, LQFP-144). The CPUs have been on the boards for longer than the 168 h that MSL 3 allows outside a dry pack, so they are baked according to J-STD-033 before each removal.
- **Resoldering report:** who performed the work and when, equipment, temperature profile, baking (temperature, duration), flux and solder, ESD protection, visual inspection after soldering with a photograph, short-circuit check.
- **Number of thermal cycles** of each CPU including the original assembly (1 reflow during board production).
- **Board labels:** the laser labels PCB1 and PCB2 must not be covered.
- **Handover and acceptance:** photographs of the boards before handover and after acceptance, signatures of both parties.
- **Prohibition:** the company does not program or run anything on the boards.

Results and conclusion

Filled in after step 7, or after the step at which the test ends, from the configuration results in the records (evaluate.txt and record.md).

Step	PCB1 + CPU	Result	PCB2 + CPU	Result
Step 1	PCB1 + CPU-X	[to be filled in]	PCB2 + CPU-Y	[to be filled in]
Step 6	PCB1 + CPU-Y	[to be filled in]	PCB2 + CPU-X	[to be filled in]
Step 7	PCB1 + CPU-X	[to be filled in]	PCB2 + CPU-Y	[to be filled in]

The result is FAIL, PASS or INCONSISTENT according to the definition of a configuration result; "not measured" if the test ended earlier.

Measured conditions in steps 1, 6 and 7 (range over all configurations): VDDIN [to be filled in], VDDCORE [to be filled in], ambient temperature [to be filled in].

Conclusion according to the interpretation table: [to be filled in]

Deviations from the procedure: [to be filled in: none, or a list with references to the configuration records]

Role	Name	Date	Signature
Performed by	Pavel Mikšíček	[to be filled in]	
Checked by	Milan Holinka	[to be filled in]	